

An Infinite Pellian Family of Integer Solutions to

$$y^2 + x^2y + z^2x + 1 = 0$$

Abstract

We give a completely explicit infinite family of integer solutions to

$$y^2 + x^2y + z^2x + 1 = 0.$$

The construction fixes $x = -145$ and reduces the equation to a Pell-type conic

$$S^2 - 580Z^2 = 21025^2 - 4.$$

A single integral point on this conic, together with the Pell unit $289 + 12\sqrt{580}$ of norm 1, generates infinitely many further integral points. A parity argument then converts all of them back to integer solutions of the original equation.

1 Statement of the result

The purpose of this note is not to classify all integer solutions. Instead, we prove the following unconditional statement, which is enough to settle the “finite versus infinite” alternative for the given Diophantine equation.

Theorem 1.1. *The Diophantine equation*

$$y^2 + x^2y + z^2x + 1 = 0 \tag{1.1}$$

has infinitely many integer solutions $(x, y, z) \in \mathbb{Z}^3$. More precisely, let

$$S_0 = 39551, \quad Z_0 = 1391, \tag{1.2}$$

and define sequences $(S_n)_{n \geq 0}$ and $(Z_n)_{n \geq 0}$ recursively by

$$\begin{aligned} S_{n+1} &= 289S_n + 6960Z_n, \\ Z_{n+1} &= 12S_n + 289Z_n. \end{aligned} \tag{1.3}$$

Then, for every $n \geq 0$,

$$(x_n, y_n, z_n) = \left(-145, \frac{S_n - 21025}{2}, Z_n \right) \tag{1.4}$$

is an integer solution of (1.1). The solutions obtained in this way are pairwise distinct. Since (1.1) contains z only through z^2 , we also get the companion solutions

$$\left(-145, \frac{S_n - 21025}{2}, -Z_n \right) \quad (n \geq 0). \tag{1.5}$$

The first few solutions produced by (1.4) are

$$(-145, 9263, 1391), \quad (-145, 10545287, 876611), \quad (-145, 6101221823, 506679767).$$

2 Reduction to a Pell-type conic

We first record the exact equivalence between the original equation with $x = -145$ and a quadratic Pell-type equation.

Lemma 2.1 (Completing the square). *Let $z \in \mathbb{Z}$ and $y \in \mathbb{Z}$. Put*

$$S = 2y + 21025. \quad (2.1)$$

Then

$$y^2 + 21025y - 145z^2 + 1 = 0 \quad (2.2)$$

holds if and only if

$$S^2 - 580z^2 = 21025^2 - 4. \quad (2.3)$$

Conversely, if $S, z \in \mathbb{Z}$ satisfy (2.3) and S is odd, then

$$y = \frac{S - 21025}{2} \quad (2.4)$$

is an integer satisfying (2.2).

Proof. Starting from (2.1), we have

$$S^2 = (2y + 21025)^2 = 4y^2 + 4 \cdot 21025y + 21025^2.$$

Therefore

$$\begin{aligned} S^2 - 580z^2 - (21025^2 - 4) &= 4y^2 + 4 \cdot 21025y + 21025^2 - 580z^2 - 21025^2 + 4 \\ &= 4(y^2 + 21025y - 145z^2 + 1). \end{aligned}$$

Hence (2.2) is equivalent to (2.3).

For the converse, assume $S, z \in \mathbb{Z}$ satisfy (2.3) and that S is odd. Since 21025 is also odd, $S - 21025$ is even, so $y = (S - 21025)/2$ is an integer. Substituting $S = 2y + 21025$ into (2.3) and using the first part of the proof gives (2.2). $\square$

Because $x = -145$ gives $x^2 = 21025$, equation (2.2) is exactly the original equation (1.1) after x has been fixed equal to -145 :

$$y^2 + x^2y + z^2x + 1 = y^2 + 21025y - 145z^2 + 1.$$

Thus, to prove Theorem 1.1, it remains to construct infinitely many integer pairs (S, z) satisfying (2.3), with S odd.

3 A norm-preserving Pell transformation

Let

$$Q(S, Z) = S^2 - 580Z^2. \quad (3.1)$$

The Pell-type equation (2.3) is simply

$$Q(S, Z) = 21025^2 - 4.$$

We shall use a norm-one automorphism of the binary quadratic form Q .

Lemma 3.1 (The Pell unit). *The integers*

$$U = 289, \quad V = 12$$

satisfy

$$U^2 - 580V^2 = 1. \tag{3.2}$$

Equivalently,

$$289^2 - 580 \cdot 12^2 = 1.$$

Proof. This is a direct calculation:

$$289^2 = 83521, \quad 580 \cdot 12^2 = 580 \cdot 144 = 83520,$$

$$\text{so } 289^2 - 580 \cdot 12^2 = 83521 - 83520 = 1. \quad \square$$

Lemma 3.2 (Norm preservation). *For every pair $(S, Z) \in \mathbb{Z}^2$, define*

$$T(S, Z) = (S', Z') = (289S + 6960Z, 12S + 289Z). \tag{3.3}$$

Then

$$Q(S', Z') = Q(S, Z). \tag{3.4}$$

In particular, if (S, Z) solves

$$S^2 - 580Z^2 = 21025^2 - 4,$$

then $T(S, Z)$ also solves the same equation.

Proof. We prove a slightly more general identity. Suppose U, V are integers and put

$$S' = US + 580VZ, \quad Z' = VS + UZ.$$

Then

$$\begin{aligned} S'^2 - 580Z'^2 &= (US + 580VZ)^2 - 580(VS + UZ)^2 \\ &= U^2S^2 + 2 \cdot 580UVSZ + 580^2V^2Z^2 \\ &\quad - 580V^2S^2 - 2 \cdot 580UVSZ - 580U^2Z^2 \\ &= (U^2 - 580V^2)S^2 - 580(U^2 - 580V^2)Z^2 \\ &= (U^2 - 580V^2)(S^2 - 580Z^2). \end{aligned}$$

Taking $U = 289$ and $V = 12$, Lemma 3.1 gives $U^2 - 580V^2 = 1$. Hence

$$S'^2 - 580Z'^2 = S^2 - 580Z^2.$$

Since $580V = 580 \cdot 12 = 6960$, this is exactly the transformation (3.3). The final assertion follows immediately. $\square$

The transformation (3.3) is the same as multiplication by the unit $289 + 12\sqrt{580}$ in the quadratic order $\mathbb{Z}[\sqrt{580}]$:

$$S' + Z'\sqrt{580} = (S + Z\sqrt{580})(289 + 12\sqrt{580}).$$

The proof above expands this identity in elementary integer arithmetic, so no algebraic number theory is being used as a black box.

4 The initial point and the infinite orbit

We now verify the initial point on the Pell-type conic and prove that its orbit under the transformation from Lemma 3.2 gives infinitely many admissible points.

Lemma 4.1 (The initial point). *The pair*

$$(S_0, Z_0) = (39551, 1391)$$

satisfies

$$S_0^2 - 580Z_0^2 = 21025^2 - 4. \quad (4.1)$$

Proof. Again this is direct arithmetic. We compute

$$39551^2 = 1564281601, \quad 1391^2 = 1934881,$$

and hence

$$580 \cdot 1391^2 = 580 \cdot 1934881 = 1122230980.$$

Therefore

$$39551^2 - 580 \cdot 1391^2 = 1564281601 - 1122230980 = 442050621.$$

On the other hand,

$$21025^2 = 442050625,$$

so

$$21025^2 - 4 = 442050625 - 4 = 442050621.$$

The two values agree, proving (4.1). $\square$

Proposition 4.2 (The recursively generated conic points). *Let $(S_n, Z_n)_{n \geq 0}$ be defined by (1.2) and (1.3). Then, for every $n \geq 0$,*

$$S_n^2 - 580Z_n^2 = 21025^2 - 4. \quad (4.2)$$

Moreover, every S_n is odd, and the sequence $(S_n)_{n \geq 0}$ is strictly increasing. In particular, the pairs (S_n, Z_n) are pairwise distinct.

Proof. First, Lemma 4.1 proves (4.2) for $n = 0$. If (4.2) holds for some n , then (1.3) says precisely that

$$(S_{n+1}, Z_{n+1}) = T(S_n, Z_n),$$

where T is the transformation from Lemma 3.2. Therefore Lemma 3.2 gives

$$S_{n+1}^2 - 580Z_{n+1}^2 = S_n^2 - 580Z_n^2 = 21025^2 - 4.$$

By induction, (4.2) holds for all $n \geq 0$.

Next we prove the parity claim. Since $S_0 = 39551$ is odd and

$$S_{n+1} = 289S_n + 6960Z_n,$$

we have, modulo 2,

$$S_{n+1} \equiv S_n \pmod{2},$$

because $289 \equiv 1 \pmod{2}$ and $6960 \equiv 0 \pmod{2}$. Thus all S_n are odd.

It remains to prove strict increase. We first show that $S_n > 0$ and $Z_n > 0$ for all $n \geq 0$. This is true at $n = 0$ because $39551 > 0$ and $1391 > 0$. If $S_n > 0$ and $Z_n > 0$, then the recurrence gives

$$S_{n+1} = 289S_n + 6960Z_n > 0, \quad Z_{n+1} = 12S_n + 289Z_n > 0.$$

Hence S_n, Z_n are positive for all n . Finally,

$$S_{n+1} - S_n = 288S_n + 6960Z_n > 0,$$

so $S_{n+1} > S_n$ for every $n \geq 0$. Therefore the sequence (S_n) is strictly increasing, and the pairs (S_n, Z_n) are pairwise distinct. $\square$

5 Proof of the main theorem

We now assemble the preceding lemmas.

Proof of Theorem 1.1. For each $n \geq 0$, Proposition 4.2 gives

$$S_n^2 - 580Z_n^2 = 21025^2 - 4,$$

and it also gives that S_n is odd. Hence Lemma 2.1 applies with $z = Z_n$ and

$$y_n = \frac{S_n - 21025}{2}.$$

Because both S_n and 21025 are odd, y_n is an integer. Lemma 2.1 then gives

$$y_n^2 + 21025y_n - 145Z_n^2 + 1 = 0.$$

Since $(-145)^2 = 21025$, this is the same as

$$y_n^2 + (-145)^2 y_n + Z_n^2(-145) + 1 = 0.$$

Thus

$$(x_n, y_n, z_n) = \left(-145, \frac{S_n - 21025}{2}, Z_n \right)$$

satisfies the original equation (1.1).

By Proposition 4.2, the sequence (S_n) is strictly increasing. Therefore the corresponding integers

$$y_n = \frac{S_n - 21025}{2}$$

are also strictly increasing. Consequently the triples in (1.4) are pairwise distinct. This gives infinitely many integer solutions.

Finally, equation (1.1) depends on z only through z^2 . Therefore, whenever $(-145, y_n, Z_n)$ is a solution, so is $(-145, y_n, -Z_n)$. This proves (1.5) as well. $\square$

6 Direct check of the first solution

For completeness, we directly verify the first solution without appealing to the general construction. When $n = 0$,

$$y_0 = \frac{39551 - 21025}{2} = 9263, \quad z_0 = 1391.$$

Then

$$\begin{aligned} y_0^2 + (-145)^2 y_0 + z_0^2(-145) + 1 &= 9263^2 + 21025 \cdot 9263 - 145 \cdot 1391^2 + 1 \\ &= 85803169 + 194754575 - 280557745 + 1 \\ &= 0. \end{aligned}$$

This agrees with the theorem and illustrates the mechanism of the construction.

Remark 6.1. *The proof is entirely constructive. The recurrence (1.3) provides a deterministic algorithm that outputs a new integer solution from the previous one. No unproved assumption, search result, or classification theorem is used.*